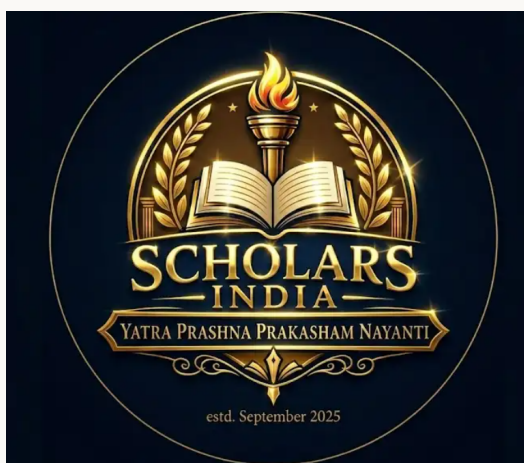


THE PIGEONHOLE PRINCIPLE

Elegant Applications in Olympiad Mathematics

Scholars India

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About the Author

Aryan Goutam

Aryan Goutam is an Olympiad mathematician, problem composer, and founder of the *Drayan Mathematical Series*. His interest in mathematics lies in the pursuit of elegant ideas—solutions that reveal unexpected structure through simple yet powerful reasoning.

Having spent years exploring competition mathematics, he developed a deep appreciation for the creativity behind problem solving and the beauty of rigorous mathematical argument. Beyond solving problems, he enjoys crafting original Olympiad-style questions and presenting mathematical ideas in a clear, engaging, and accessible manner.

The *Drayan Mathematical Series* was created with the belief that mathematics should be experienced as a creative and intellectual adventure rather than a collection of formulas and techniques. Through these handouts, he hopes to inspire students to think deeply, write carefully, and discover the elegance that lies at the heart of mathematics.

Areas of Interest:

Number Theory • Algebra • Combinatorics • Problem Creation

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Abstract

Few principles in mathematics demonstrate the power of simple ideas as strikingly as the Pigeonhole Principle. What begins as an almost obvious observation quickly evolves into a versatile method capable of revealing order within apparent chaos. From proving the inevitability of coincidences to establishing deep existence results, its applications span a remarkable range of mathematical disciplines.

This handout presents a guided exploration of the principle through carefully selected examples, classical Olympiad problems, and original exercises. The goal is not merely to master a technique, but to cultivate the habit of identifying hidden structures that make elegant solutions possible.

Target Audience: Beginner to Advanced Olympiad students.

Prerequisites: Elementary combinatorics and number theory.

Difficulty: Intermediate.

Estimated Reading Time: 50–80 minutes.

subsection*Books and Written References

Resource	Level	Commentary
<i>The Pigeon-Hole Principle</i> — Valentin Goranko	Int.	Focused treatment with numerous Olympiad applications.
<i>Pigeons, Holes, and the Pigeonhole Principle</i> — Berkeley Math Circle	Beg.–Int.	Excellent intuition-building examples.
<i>Pigeonhole Principle</i> — University of Washington Math Circle	Beg.–Int.	Clear exposition with well-structured exercises.
<i>Pigeonhole Principle</i> — UCLA Math Circle	Int.–Adv.	Strong collection of Olympiad-style problems.
<i>Problem-Solving Strategies</i> — Arthur Engel	Adv.	A classic Olympiad text with elegant combinatorial arguments.
<i>Intermediate Counting & Probability</i> — David Patrick (AoPS)	Int.–Adv.	Outstanding exposition and problem sets.

Video Lectures and Online Learning

Lecture	Level	Commentary
<i>Pigeonhole Principle (Part 1)</i> — Vedantu Olympiad School	Beg.–Int.	Complete introduction to the classical and generalized Pigeonhole Principle with IOQM-oriented examples.
<i>Pigeonhole Principle (Part 2)</i> — Vedantu Olympiad School	Int.–Adv.	Advanced applications and Olympiad-level problem solving.
<i>Pigeonhole Principle Lecture</i> — Prashant Jain Sir (Unacademy)	Int.	Concise treatment with contest-focused examples and techniques.
<i>Olympiad Combinatorics Lectures</i> — Prashant Jain Sir	Int.–Adv.	One of the most respected collections of Indian Olympiad training lectures.
<i>The Pigeonhole Principle</i> — MIT OpenCourseWare	Adv.	Rigorous treatment emphasizing proofs and non-standard applications.
<i>Problem Solving with the Pigeonhole Principle</i> — Michael Penn	Adv.	Strong collection of Olympiad and undergraduate competition problems.
<i>The Pigeonhole Principle: 7 Gorgeous Proofs</i> — Mathologer	Adv.	Beautiful applications illustrating the depth and versatility of the principle.

1 Standard Problems

Problem

Prove that in any group of 13 people, at least two share the same birth month.

Solution

Idea. Direct application of the pigeonhole principle.

Solution.

There are 12 possible birth months (pigeonholes) and 13 people (pigeons). By the pigeonhole principle, at least one month must contain at least two people. Hence, at least two people share the same birth month.

Remark

More generally, among $12k + 1$ people, some month is shared by at least $k + 1$ people.

Problem

Five points are placed inside an equilateral triangle of side length 2. Show that at least two are at distance at most 1.

Solution

Idea. Partition the domain into smaller regions of controlled diameter.

Solution.

Join the midpoints of the sides to divide the large triangle into four smaller equilateral triangles of side length 1. By the pigeonhole principle, with five points and four regions, at least one small triangle contains two points. The diameter of each small triangle is 1, so those two points are at distance at most 1.

2 Problems by Aryan Goutam

Problem

Let S be a set of $n^2 + 1$ distinct positive integers. Prove that there exist two nonempty disjoint subsets $A, B \subseteq S$ such that $\sum A = \sum B$.

Motivation. This problem shows how abundance forces equality in combinatorial structures via the pigeonhole principle on subset sums.

Solution

Idea. Consider the power set and apply the pigeonhole principle on subset sums.

Solution.

The set S has 2^{n^2+1} subsets. The possible subset sums range from 0 to the total sum of elements in S . By the pigeonhole principle there must exist two distinct subsets X and Y with equal sums. Let $A = X \setminus Y$ and $B = Y \setminus X$. Then A and B are disjoint and $\sum A = \sum B$.

Remark

This result admits various strengthenings, including constraints on subset sizes or modular analogues.

Closing Remarks

Recommended Next Topics

- Invariants in Olympiad Mathematics
- Functional Equations
- Vieta Jumping
- Power of a Point

“The mathematician’s patterns, like the painter’s or the poet’s, must be beautiful; the ideas, like the colours or the words, must fit together in a harmonious way.”

— G. H. Hardy